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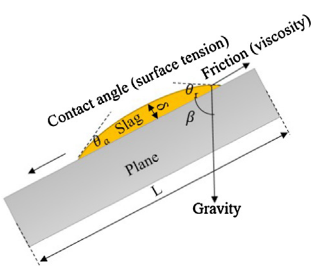
Establishing a novel and yet simple methodology based on the use of modified inclined plane (M-IP) for high-temperature slag viscosity measurement

Baiqian Dai[a], Xiaojiang Wu[a,b], Lian Zhang[a,*]

[a] *Department of Chemical Engineering, Monash University, Wellington Road, Clayton, Victoria 3800, Australia*
[b] *R&D Division, Shanghai Boiler Works Co. Ltd, Minhang, Shanghai 200245, China*

GRAPHICAL ABSTRACT



(b) Schematic of slag falling film model

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ABSTRACT

In this study, a simple and yet novel methodology based on the use of modified inclined plane (M-IP) has been established and validated to determine the high-temperature slag viscosity. Seven synthetic standard ash samples have been tested on the inclined corundum plane, under the conditions of reducing environment of 1% CO in nitrogen, 1100 °C–1400 °C, 25°–90° and different duration time of 10–40 min. The slag mass was varied from 100 *mg* to 400 *mg*. A multiple linear regression fitting was conducted to establish an empirical equation to predict slag viscosity (η) based on the slag travel length per unit mass (L') and the inclination angle of the plane ($cos\beta$). As has been fully validated, the new method is simple and requires a maximum of 200 *mg* ash for the slagging test. By measuring only two temperature points, an Arrhenius equation can be established to predict the travel length at any of the intermediate temperatures. Both the inclination angle and the reciprocal of slag viscosity affect the slag travel length in an intertwined manner, by following a linear relationship between the algorithm of slag travel length per unit mass, $cos\beta$ and the logarithm of slag viscosity, as per the equation of $ln\mu = 3.282281cos\beta - 1.882827lnL' + 7.397108$. The large correlation coefficient value and the normal quantile plot supported the high reliability of this equation. However, this empirical equation is limited to an upper temperature of 1400 °C and a maximum viscosity of 17.9 *Pa·s*. Any larger viscosity would only be measured qualitatively, whilst it is sufficient for a comparison with the critical viscosity 25 *Pa·s* from the perspective of a quick screening of solid fuel for entrained-flow gasifier or cyclone furnace.

* Corresponding author.
E-mail address: lian.zhang@monash.edu (L. Zhang).

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Nomenclature

A_A	pre-exponential factor for Arrhenius equation [$g\,cm^{-2}\,atm^{-1}\,s^{-1}$]
E_A	activation energy for Arrhenius equation [$kj\,mol^{-1}$]
L	length of the plane [mm]
ρ	bulk density [$g\,cm^{-3}$]
g	gravity constant [−]
β	inclination angle [°]
v_z	velocity in z direction [$m\,s^{-1}$]
τ_{xz}	molecular stress tensor component on surface [N]
θ_a	advancing angle [°]
θ_r	receding angle [°]
μ	dynamic viscosity [*Pa·s*]
η	viscosity [*Pa·s*]
δ	thickness of slag film [mm]
n	flow behaviour index [−]
R	gas constant [−]
L'	slag travel length per unit mass [$mm\,g^{-1}$]
K	consistency index [−]

1. Introduction

Slag viscosity has implications in glass production, geophysical phenomena, metallurgical processes, fly ash vitrification, combustion and gasification. The viscosity determines the performance of slagging, which minimises the volume and drain of the ash from the high temperature process [1]. The formation of slag critically affects the heat transfer and the stability of the combustion and gasification processes [2]. The viscosity of a slag and its change with temperature are vital as they will determine the extent of motion taking place at the interface and affect the successful operation of high-temperature systems including gasifier, cyclone furnace and metallurgy furnace [3].

Viscosity is the ability to resist the movement of one layer of molecules over another when a stress is applied. It is a measurement of the internal friction of a fluid, which significantly depends upon structure, physical properties and temperature. For a coal slag sample, its viscosity is normally dominated by its silicate structure [4]. An acidic slag with a large quantity of polymerised silicate ions is highly viscous, whereas a basic slag with small de-polymerised silicates is much more like a fluid [4]. To date, a wide range of techniques has been developed for the viscosity measurement, as summarised in Table 1 [3–6]. The capillary method is normally used in a Shiraishi Viscometer, detecting the parallel plate sample, height and time, and covers a large detecting range from 100 to 10^{11} *Pa·s* [5]. However, it is restricted to the highly viscous fluids and impractical at high temperatures (> 1500 °C). The falling body viscometer determines the viscosity by measuring the time for a slag bob to drag through a measured distance. The measurement range varies from 3 to 100,000 *Pa·s* [4,5]. Its limitation is that it requires a long (> 200 *mm*) and uniform high temperature zone, and accurate operations. The oscillating viscometer measures the logarithm decrement of amplitude of twisting. It is normally applicable for low viscosity fluids (< 0.1 *Pa·s*), and requires melt density to determine the viscosity according to the Roscoe Equation [5,6]. The rotating cylinder viscometer covers a viscosity range from 0.01 to 100 *Pa·s*, which is a desirable range for most coal slags. Therefore, it has been widely used for measuring the viscosity of coal slags [3]. However, due to the difficulty in torque measurement, it requires very accurate vertical alignment and skilled operators. The uncertainty in the measurements for many slags is normally higher than ± 20%, due to the changes on slag composition during the measurement and the systematic errors from the equipment [6]. The particle precipitation and solidification in the slag also affect the accuracy of the results significantly.

The difficulty and high cost of viscosity measurement for slags has led to the establishment of a large number of empirical models based on the slag chemical composition. Table 2 summarises the most common models for slag prediction [7–14]. The Urbain Model is one of the most widely used slag viscosity models, which is based on the CaO-Al_2O_3-SiO_2 system. The slag constituents are classified into three categories, glass formers, modifiers and amphoteric [7]. Kondratiev and Jak modified the Urbain viscosity model to include the multi-component slags [8]. The four-component system SiO_2-Al_2O_3-CaO-"FeO" slag is used to describe the viscosity behaviour over the whole composition range instead of CaO-Al_2O_3-SiO_2 system in Urbain Model. The Ribound model was initially developed to estimate the viscosity of mould powders in the SiO_2-CaO-Al_2O_3-CaF_2-Na_2O system [9]. It has been extended to estimate the viscosity for other slags. However, it fails to differentiate between the various cations [9]. Mills and Sridhar have developed the National Physical laboratory (NPL) model [10]. It relates the viscosity of slag to the structure through optical basicity correcting [10,11]. However, it is only applicable for standard glasses and is inaccurate for slags. The Lida model is based on the Arrhenius type of equation, where the network structure of a slag is taken into account by using the basicity index [12,13]. However, it is implausible to apply the Lida model to systems when there is no experimental data for an accurate calibration. The high accuracy claimed by this model only comes from its calibration with experimental data for each slag [6]. The KTH model is based on the Eyring equation rather than the Arrhenius equation [14]. It takes into account the Gibbs energy of activation for viscosity. The model is commercially available in "THERMOLSLAG" software [3,15]. However, the presence of net-forming and non-net-forming oxides in the slag, and the complicated interaction between cations and anions make the prediction using this model problematic.

There is a well-defined need for a simplified test to measure the viscosities for quality assurance of supplies on a routine basis. Mills investigated a simple test for the slag viscosity measurement by using an inclined plane [16]. An ash sample was first melted into slag at high temperatures (1200 °C–1400 °C). Subsequently, the resultant molten slag was carefully poured onto an inclined plane to flow down. The inclined angle is only restricted to 9° and 23° and the viscosity range is restricted in the range of 1.5–6 *Pa·s* with an error of ± 15%. Moreover, the prior melting makes this measurement risky and uncertain in operation. Any cooling during the transferring could affect the accuracy of the measurement.

This paper aims to modify the inclined plane methodology to create a fast and simple high-temperature methodology for slag viscosity measurement, namely modified inclined plane (M-IP) method. Its novelty includes, 1) a prior melting is not necessary, 2) a small amount of ash sample up to 200 *mg* is sufficient, 3) the methodology is

Table 1
Summary of the existing viscosity measurement methods [3–6].

Methods	Range. (*Pa·s*)	Specification	Limitation	Ref
Capillary	10^2–10^{11}	Rotating plate-torque, Parallel plate-sample height-time	Unpractical for high temperature	[5]
Falling body	3–10^5	Time for bob to fall/drag through measured distance	Long and uniform hot zone required	[4,5]
Oscillating	10^{-4}–0.1	Measures log decrement of amplitude of twisting	Applicable for low viscosity de-polymerised slag	[5,6]
Rotating cylinder	0.1–100	Measuring the torque when rotating the bob or crucible	Accurate alignment and experienced operation required	[3]

Table 2
Summary of the existing viscosity prediction models [7–14].

Models	Slag type	Dependence relation	Comments	Ref
Urbain	Various	$\eta = A_A e^{\frac{E_A}{RT}}$	Based on CaO-Al_2O_3-SiO_2 system, Valid only for limited composition-temperature ranges	[7]
Modified Urbain	Coal, Synthetic	$\eta = ATe^{\frac{1000B}{T}}$	Based on CaO-Al_2O_3-SiO_2-FeO four component system, Valid for whole composition range within four components	[8]
Riboud	Mould powder	$\eta = ATe^{\frac{B}{T}}$	SiO_2-CaO-Al_2O_3-CaF_2-Na_2O system, Fails to differentiate between the various cations.	[9]
NPL	Industrial slags and mold fluxes	$\eta = Ae^{\frac{B}{T}}$	Optimized using a limited set of experimental data, Accurate for standard glasses, but less accurate for slags.	[10,11]
Lida	Mold fluxes and metallurgical slags	$\eta = A\eta_0 e^{\frac{E}{B_i}}$	Basicity index value required, and experimental data calibration required for accurate prediction	[12,13]
KTH	Metallurgical slags	$\eta = \frac{hN_A\rho}{M} A_A e^{\frac{G}{RT}}$	Coefficients are not published	[14]

independent on slag composition, and 4) the methodology measures the travel length of the whole slag regardless of it behaving as a Newtonian fluid or an non-Newtonian fluid. Therefore, it considers the effects from particle precipitation and solidification as well. Based on the M-IP method, an empirical equation was further established to predict the slag viscosity based on the inclined angle and the travel path per unit mass of the slag. To validate the empirical equation, seven typical standard ash samples were synthesised and tested in a temperature range of 1100–1400 °C under a reducing environment of 1% CO balanced in nitrogen. The key factors including temperature, slag mass, residence time and inclined angle were assessed systematically. The relationship between the slag travel length, inclined angle and the slag viscosity has been elaborated by using a modified falling film model. The reliability of the established equation was validated by the large correlation coefficient and normal quantile plot. This simple and yet novel method is expected to be used for the calculation of the viscosity of a slag irrespective of its composition and the feedstock.

2. Experimental and theory

2.1. Experimental

2.1.1. Ash sample preparation

Seven standard ash samples, covering a viscosity from 1.01 to 212 *Pa·s* at 1400 °C [17] were synthesised by using Sigma-Aldrich analytical grade oxides, including aluminium oxide (Al_2O_3), silicon oxide (SiO_2), calcium oxide (CaO) and iron oxide (Fe_2O_3). The purity for all the oxides is 99.99% and the size is less than 10 μm. The oxides were blended using a dual mixer (8000D SPEX Sample Prep) for complete mixing and homogenisation. Table 3 provides the ash compositions for the synthetic standard ash samples. From ash 1–7, the viscosity at 1400 °C follows an ascending trend, while the SiO_2 content shows an increasing trend. The lowest viscosity ash sample 1 consists of the least SiO_2 content by 38.6% and the highest CaO content by 30.4%. The synthetic ash samples were measured accurately before being loaded onto a corundum plane, which was made of Al_2O_3 with a purity of 99.9%. A square plastic mould with a size of 10 *mm (wide)* × 10 *mm (length)* × 2 *mm (depth)* was employed to shape the ash sample. The ash powders were pressed by hand to ensure a stable adherence of the powders onto the plate. For each run, the ash sample was located in the same position of the plane by accurately aligning the mould and plane position.

2.1.2. Experimental procedure

The experimental facility and procedures were described in our previous studies [18,19]. As shown in Fig. 1, an electrically heated high temperature horizontal reactor was used. The slagging took place inside a corundum tube with an inner diameter of 70 *mm* and a length of 1200 *mm*. The ash-laden corundum plane was placed and inclined to a certain angle in a ceramic sample holder. In this paper, four inclinations, including 25°, 45°, 60° and 90° were tested, as shown in Fig. 1. For each test, the furnace was pre-heated to a set temperature under the protection of pure argon. Fig. 2 indicates the temperature profile for the reactor. Once the set temperature reached, the ash-laden corundum plane was pushed into the middle of the furnace, where the temperature has been calibrated and is uniform. Note that, the sample was held at three different temperatures/stages for around 1 min per stage to make sure that its temperature is identical with the furnace temperature. The reducing gas of 1 *vol.%* CO in N_2 was simultaneously switched on once the sample was injected into the reactor. Its flow rate maintained at ~1 L/min, and the exposure time was maximum 40 min. After the exposure test, the ash holder was quickly pulled out from the hot middle zone and moved to the cold end of the furnace, whereas the gas was changed promptly back to pure argon with a large flow rate of 10 L/min to quench the hot slag. Subsequently, the sample was taken out from the furnace and was quenched in a dry ice box to avoid oxidation and secondary reactions.

2.2. Hypothetical theory

Most metallic melts and molten slags are Newtonian fluids where the viscosities depend on the shear rate. Therefore, the viscosity of a molten slag is usually defined according to Newton's law as the proportionality constant between the shear stress and the velocity gradient normal to the shear stress [15,20]. However, both Newtonian and non-Newtonian behaviours have been observed in the ash slagging systems

Table 3
Compositions for synthetic standard ash samples [17].

Sample No	SiO_2	Al_2O_3	CaO	Fe_2O_3	Na_2O	MgO	SO_3	K_2O	TiO_2	Viscosity at 1400 °C
1	38.6	17.9	30.4	10.5	0.06	0.13	0.54	0.01	0.01	1.01
2	44.6	25.2	19.5	11.2	0.12	0.09	0.37	0.01	0.01	4.96
3	47.8	25.2	14.7	11.6	0.13	0.07	0.29	0	0.02	11.1
4	56	15.6	20.1	5.79	0.12	0.15	0.37	0	0.02	17.9
5	55.5	25	9.8	11.2	0.15	0.06	0.22	0.01	0.02	46
6	60.5	19.6	9.1	10.7	0.14	0.06	0.19	0.01	0.02	102
7	60.8	20.8	10.4	5.84	0.12	0.09	0.2	0.01	0.01	212

(Ash composition in *wt*%, dry basis, Viscosity in *Pa·s*).

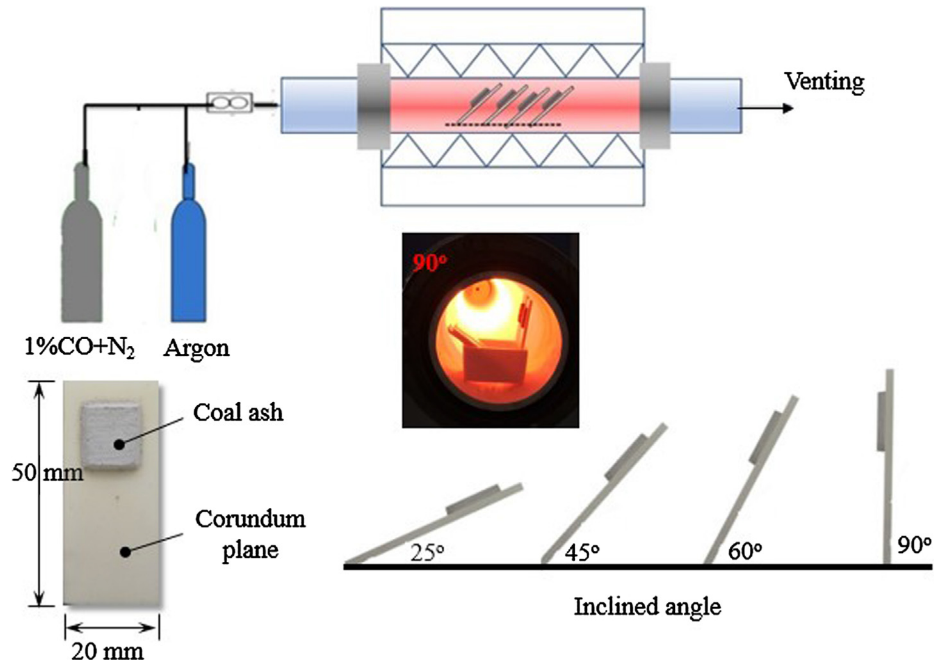


Fig. 1. The high temperature horizontal reactor and sample loading plane.

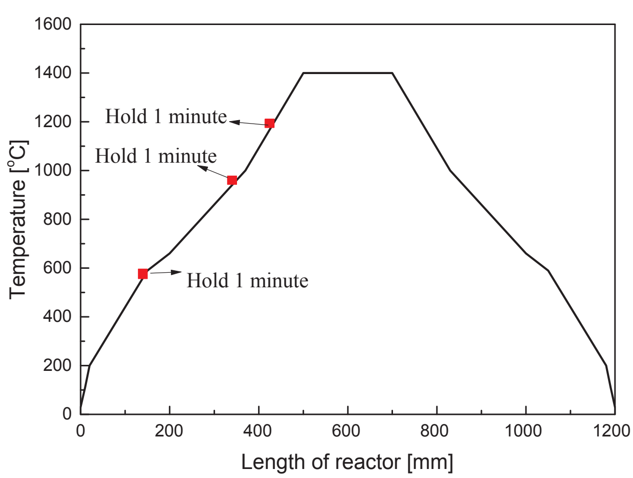


Fig. 2. Temperature profile for the high-temperature horizontal reactor and sample loading procedure.

[21]. Newtonian behaviour has been reported to account for the slags containing less than 10–40 *vol.%* solids [21]. In this study, a falling/creeping film model was used to correlate the slag flow velocity and its viscosity, based on the classical falling film theory [22]. Fig. 3 illustrates the side view of the cross-section of slag and the falling film model, where the velocity of the film is governed by the force balance between gravity, friction force and even the surface tension of the slag. Eq. (1) describes the relationship between the surface tension and contact angles, and this relation has been modelled using a variety of approaches for parallel-sided or elliptical contact lines [23].

$$\frac{F}{\gamma \mathrm{w}} = k\,(cos\theta_r - cos\theta_a) \tag{1}$$

where F is the critical force, γ is the surface tension, w is the width, k is a numerical constant that depends on the shape of the drop. The angles of θ_a and θ_r represent the advancing angle and receding angle, respectively [23]. From the optical microscopy image for the slag in Fig. 3a, the thickness of the film is less than 0.1 *mm*, and the advancing angle θ_a and the receding angle θ_r for a thin falling slag film is roughly the same. Hence, the hysteresis (H) based on the difference of the two angles in Eq. (1) is nearly zero, meaning that the surface tension is negligible.

For a falling film, its force balance at a control volume of Δx can be expressed as Eq. (2):

$$\frac{\mathrm{d}\tau_{xz}}{\mathrm{dx}} = \rho g cos\beta \tag{2}$$

Integrate Eq. (2) and use the boundary condition at x = 0, τ_{xz} = 0, the shear stress can be presented as Eq. (3):

$$\tau_{xz} = (\rho g cos\beta)x \tag{3}$$

Based on the assumption that the slag was a Newtonian fluid, it will then follow Newton's law of viscosity in Eq. (4):

$$\tau_{xz} = -\mu\frac{dv_z}{dx} \tag{4}$$

After integration of Eq. (4) by using the boundary condition at $x = \delta,\ v_z = 0$, the average velocity (v_z) over a cross-section of the film can be obtained below:

$$v_z = \frac{\rho g\delta^2 cos\beta}{3\mu} \tag{5}$$

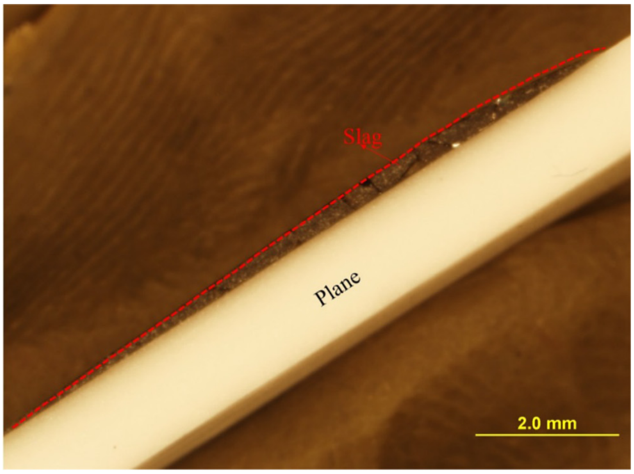
Apparently, if the slag is a Newtonians fluid, its averaged velocity (travel length divided by time) should increase proportionally to $cos\beta$ and the reciprocal of its viscosity, μ. However, if the slag is a non-Newtonian fluid, the power law Eq. (6) will apply to correlate its viscosity and the respective shear stress, instead of Eq. (4) [24,25].

$$\tau_{xz} = -\mu\left(\frac{dv_z}{dx}\right)^n \tag{6}$$

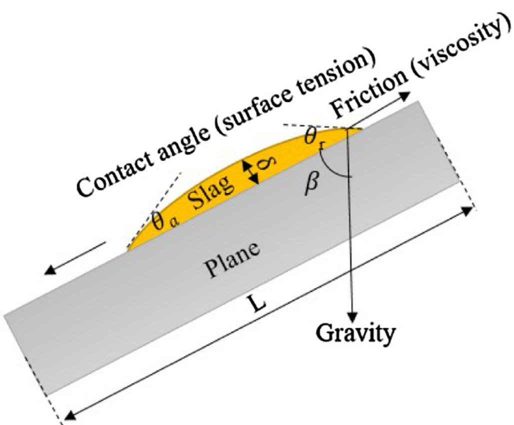
Put the Eq. (6) back to the force balance Eq. (3), and integrate Eq. (3) from the boundary condition of $x = \delta,\ v_z = 0$, the force balance of the falling film can be expressed in Eqs. (7)–(9) as below:

$$-\mu\left(\frac{dv_z}{dx}\right)^n = (\rho g cos\beta)x \tag{7}$$

$$-\mu\left(\frac{dv_z}{dx}\right) = \{(\rho g cos\beta)x\}^{1/n} \tag{8}$$



(a) Optical microscopy of cross-section of slag on the plane



(b) Schematic of slag falling film model

Fig. 3. The optical image for cross-section of slag and the slag falling film model, (a) optical microscopy of slag cross-section, (b) slag falling film model.

$$\frac{dv_z}{dx} = -\frac{(\rho g cos\beta)^{1/n}}{\mu}x^{1/n} \tag{9}$$

Further integrate the Eq. (9) from the boundary condition at $x = \delta$, $v_z = 0$, the average velocity (v_z) over a cross-section of the film for a non-Newtonian slag can be obtained below:

$$v_z = \frac{(\rho g cos\beta)^{1/n}}{(1/n + 2)\mu}\delta^{\frac{1}{n}+1} \tag{10}$$

Take the logarithm of the two sides of Eq. (10), one can establish the following linear correlation Eqs. (11) and (12) between the dependent variable, viscosity (μ) and the two independent variables, inclined angle ($cos\beta$) and the slag travel velocity (v_z). In particular, it is obvious that the two independent variables will affect the viscosity to different extents, as evident by the different coefficients (1/n and −1) in Eq. (12).

$$\ln(v_z) = \frac{1}{n}ln(cos\beta) - \ln(\mu) + \log\left(\frac{(\rho g)^{\frac{1}{n}}\delta^{\frac{1}{n}+1}}{(1/n + 2)}\right) \tag{11}$$

$$\ln(\mu) = \frac{1}{n}ln(cos\beta) - \ln(v_z) + \log\left(\frac{(\rho g)^{\frac{1}{n}}\delta^{\frac{1}{n}+1}}{(1/n + 2)}\right) \tag{12}$$

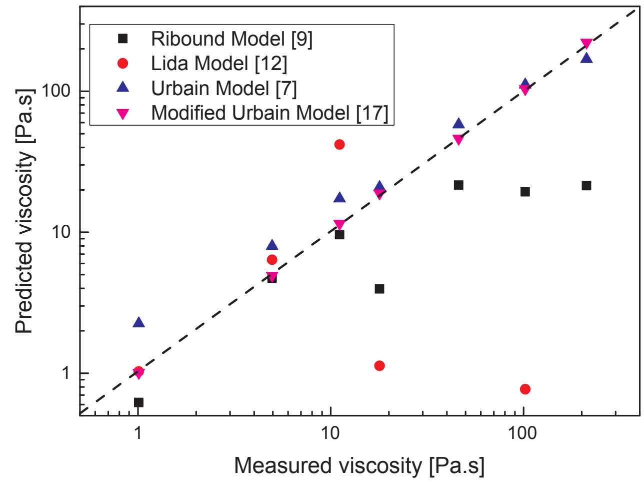


Fig. 4. The comparison of predicted and measured viscosities for the standard ash samples.

3. Results and discussion

Prior to the inclined plane slagging tests, effort was made to apply the current viscosity models to the selected seven standard ash samples. Fig. 4 shows how the calculated results match the experimental results, where the measurements were conducted at 1400 °C elsewhere [17,26]. Clearly, most of the model results fail to match with experimental results. This further demonstrates the necessity of establishing a reliable and accurate method for the slag viscosity.

3.1. Effects of mass, temperature, and residence time

The mass of ash/slag is the sole driving force for the falling film on the inclined plane. It directly affects the bulk density (ρ) of the falling film. In other words, the more ash packed into the module with a fixed volume of 10 *mm* × 10 *mm* × 2 *mm*, the larger the bulk density. In light of this, the effect of mass on the slag flow-ability was firstly examined by testing the ash sample No 2 with a viscosity of 4.96 *Pa·s*, and five different masses, 0.1 *g*, 0.2 *g*, 0.25 *g*, 0.3 *g* and 0.4 *g* were investigated. Note that, the ash exposure conditions were fixed at 1400 °C with an inclination angle of 25° for 40 min. Fig. 5 provides the slag travel length as a function of mass. It is evident that the slag with 0.1 *g* mass travels the least, because there is not sufficient ash to achieve a proper momentum. The slag travel length increases promotionally as the mass increases, and the slag with the mass of 0.4 *g* overflows and reaches the bottom of the plane. A linear trend was confirmed between the slag travel length per unit mass and its mass in the range from 0.1 *g* to 0.3 *g*. This proves the suitability of the falling film theory, by which the velocity is in proportion to the density in both Eq. (5). The use of 0.4 *g* is the only outlier here, which is simply due to the practically limited length of the plane. Therefore, the mass for all the tests was fixed at approximately 0.2 *g* hereafter. The use of less mass makes the travel length measurement less accurate, while the use of more mass could cause the overflowing when a sharp angle such as 90° is used. Hereafter, the slag travel length was further normalised as per unit mass to eliminate the effect of mass.

Secondly, the exposure temperature was tested, considering it is a critical factor determining the molten fraction of a slag and its viscosity significantly. Initially, all the ash samples were tested at temperatures from 1300 °C to 1400 °C at an internal of 50 °C. However, the highly viscous ash samples from No 5 through to No 7 did not flow at 1400 °C. Subsequently, these three samples were ruled out. The remaining four samples were tested and discussed in depth hereafter. Additionally, the No 2 with a very low viscosity of 4.96 *Pa·s* was tested in a wider temperature span from 1100 °C to 1400 °C. Fig. 6 provides the images for all

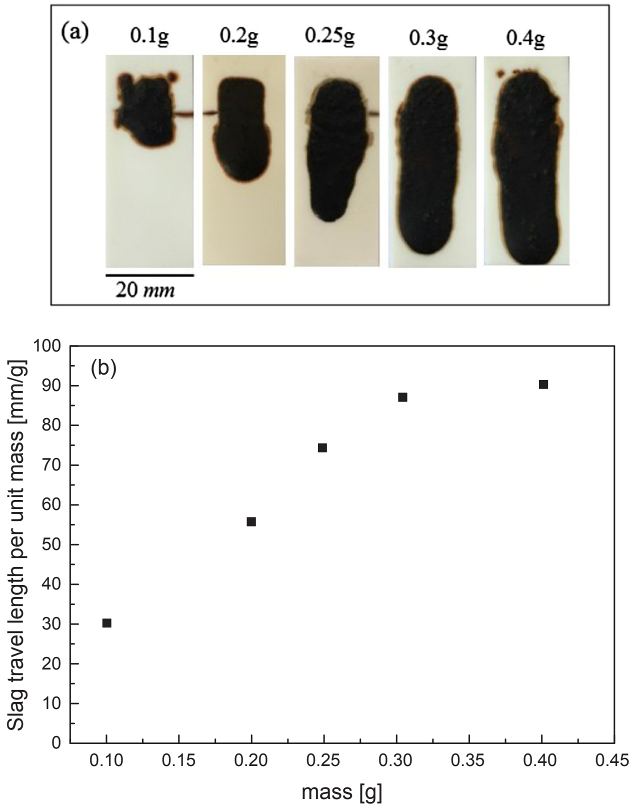


Fig. 5. The effect of ash mass on the slag flow-ability for ash 2. The exposure time is 40 min at 1400 °C with an inclination angle of 25°. (a) Images for slag flow, (b) slag travel length vs mass.

the slag samples obtained from different temperatures. For the tests here, the holding time is 40 min and the inclined angle is 25°. It is obvious that ash 2 shrinks at 1200 °C and starts to flow from 1300 °C. Ash 1 travels about 50% length of the plane at 1300 °C, while the ash 3

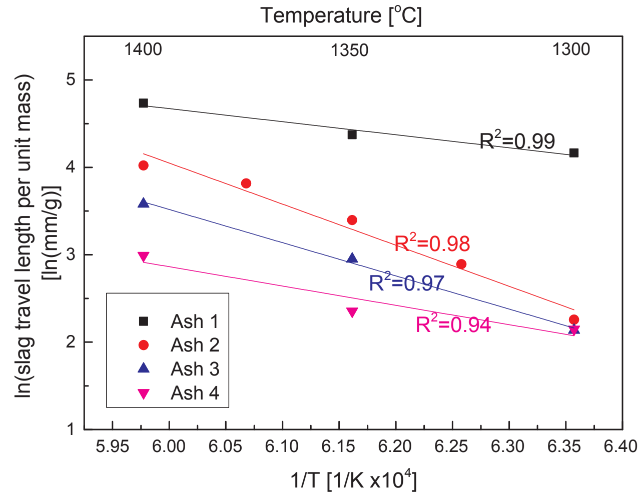


Fig. 7. The relationship between slag travel length per unit mass and temperature for four ash samples from ash 1 to ash 4.

and ash 4 only start to melt and shrink at 1300 °C. This is because the ash 1 sample has the least viscosity while No 3 and 4 ashes have the largest viscosity. As temperature increases, the slag travels more quickly for all the samples, as expected. To establish the relationship between slag travel length (*i.e.* velocity) and temperature, effort was made by taking the logarithm of the former variable for each ash and plotting it against the temperature reciprocal in Fig. 7. It is clear to see a good linear relationship achieved between the two variables, irrespective of ash sample. The reliability of the linear relationship is validated by the large correlation coefficient. Since the logarithm of the slag travel length is in negative proportion to its viscosity, as evident in Eqs. (5) and (12), one can further infer that the temperature dependence of the travel length of slag on the inclined plane follows an Arrhenius-type relationship Eq. (13), which is widely used for the temperature dependence of slag viscosity [27–29].

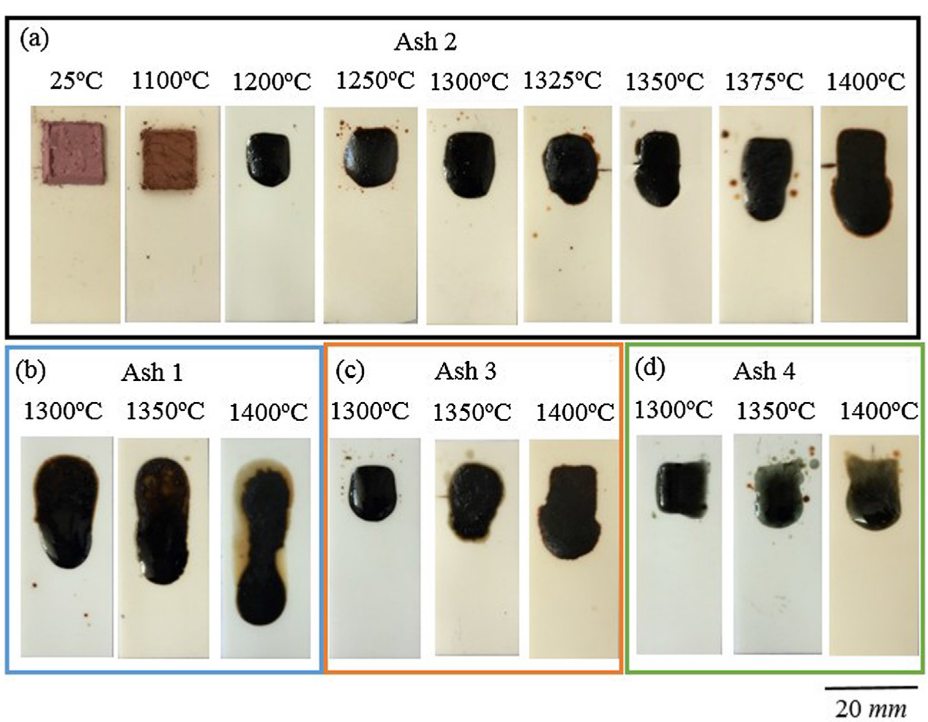


Fig. 6. Slag flow for ash samples at different temperatures. The exposure time is 40 min with an inclination of 25°, (a) ash 2, (b) ash 1, (c) ash 3 and (d) ash 4.

$$\eta = A_A e^{\frac{E_A}{RT}} \tag{13}$$

where A_A is the pre-exponential factor, R is the gas constant, E_A is the activation energy and T is temperature. It can be derived into Eq. (14) by taking a logarithm expression.

$$\ln(\eta) = \ln A_A + (E_A/R)\frac{1}{T} \tag{14}$$

Thirdly, the effect of residence time was examined by testing the top two ash samples No 1 and No 2 at 1400 °C, an inclined angle of 25° and four different duration times, 10 min, 20 min, 30 min and 40 min. The results are visualised in Fig. 8a. A linear relationship with a large correlation coefficient was confirmed between the slag travel length unit mass and the residence time for both ash samples, as evident in Fig. 8b. Each ash melts and flows instantaneously on the inclined plane. The velocity of a molten slag is also stable and constant over the exposure time, suggesting that it can be simply calculated by dividing the travel length (per unit mass) over the exposure time. This solves the velocity term (v_z) in Eqs. (5) and (12).

3.2. Effect of inclination angle

For the molten slag at high temperature, the travel length is expected to be a function of cosβ as suggested by the two Eqs. (5) and (12). Fig. 9 provides the images for four ash samples at 1400 °C with different inclination angles, 25°, 45°, 60° and 90°. Note that, the exposure time was fixed at 40 min for all the tests except three cases, ash 1 and ash 2 at 60° and ash 1 at 90°. For the former two cases, a duration time of 30 min was used, whereas for the latter case was only tested in 15 min because the slag flew too fast in these cases. Since the slag travel length is in proportion to the residence time in Fig. 8, the travel length results achieved for the three exceptional cases were normalised to 40 min. For the least viscous ash No 1, it travels three quarters of the whole plane length at 25° inclination angle over a 40 min period, whereas it travels to the end of the plate as the inclination increases to 45° over the same residence time. It travels faster when the inclination further increases to 60°, where it reaches the end of the plane by only 30 min. The slag travels more than half of the plane length within 15 min by further increasing the inclined angle to 90°. Ash 2 travels slower compared to ash 1. It travels half of the whole plane length at 25° within 40 min, and it approaches to the end of the plate by increasing the inclination to 60°. Further increasing the inclination to 90° increases the slag flow rate by approaching the end within only 30 min. The highly viscous samples, ash 3 and 4, flow much more slowly, although an obvious increment for the slag travel length can be observed when increasing the inclination. The ash 3 can travel nearly half of the plane length and ash 4 can travel more than one third of the plane length at 90° within 40 min. Finally, Fig. 10 plotted the unit travel length versus cosβ for the four ash samples. A linear correlation was confirmed for all the four ash samples, proving the suitability of the falling film theory discussed before. In addition, these findings are consistent with Mills [16].

3.3. Empirical equation for the prediction of slag viscosity

As mentioned in Section 2.2, in the case that the molten slag is a Newtonian fluid, its velocity would follow the Eq. (5) for a linear correlation with cosβ and an inverse proportion to the slag viscosity. For the majority of the exposure tests conducted in 40 min here, the velocity can be simply calculated by dividing the travel length (per unit mass) by 40 min. For each specified inclined angle, cosβ is a constant. Therefore, the travel length of a slag and the reciprocal of its viscosity should follow a linear relationship as shown in Eq. (15).

$$L' = K\eta^{-1} \tag{15}$$

where the symbol L' refers to the travel length per unit mass, and K is a constant that can be theoretically calculated by the expression of $(\rho g \delta^2 \cos\beta \Delta t)/3$, as per Eq. (5).

Fig. 11 provides the relationship between the slag travel length and reciprocal of the respective viscosity. Intriguingly, a linear correlation was not found for these two variables. Instead, a power-law relationship was observed for each inclination angle. This is a strong indicator for the non-Newtonian behaviour of the four ash samples tested here. Therefore, the Eq. (12) should apply, in which the slag velocity (*i.e.* travel length) is a function of both inclined angle and the slag viscosity, but at a different power for these two independent variables. Bearing this in mind, the travel lengths of the four slags and their respective viscosities were taken in the logarithm form, and plotted in Fig. 12. Regardless of the inclined angle, a linear correlation was confirmed for each inclination angle. Moreover, the different angles bear a different slope, indicating the intertwined effect from the two independent variables, inclination angle and slag viscosity. This proves the non-Newtonian behaviour of slag and the suitability of Eq. (12).

Using the unit travel length (L') and exposure duration to replace the velocity (v_z) in Eq. (12), one can achieve a generic Eq. (16) for the expression of slag viscosity versus travel length and inclined angle.

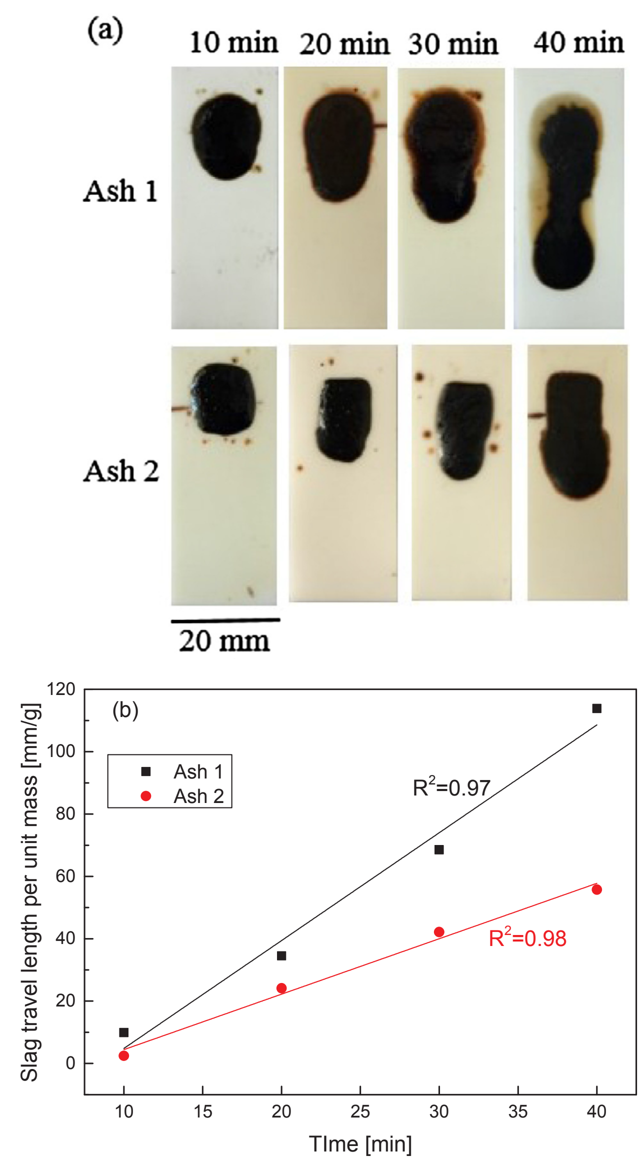


Fig. 8. The effect of residence time on slag flow-ability for ash 1 and 2. The temperature is 1400 °C with inclination of 25°. (a) Images for slag in different residence time, (b) relationship of slag travel length and residence time.

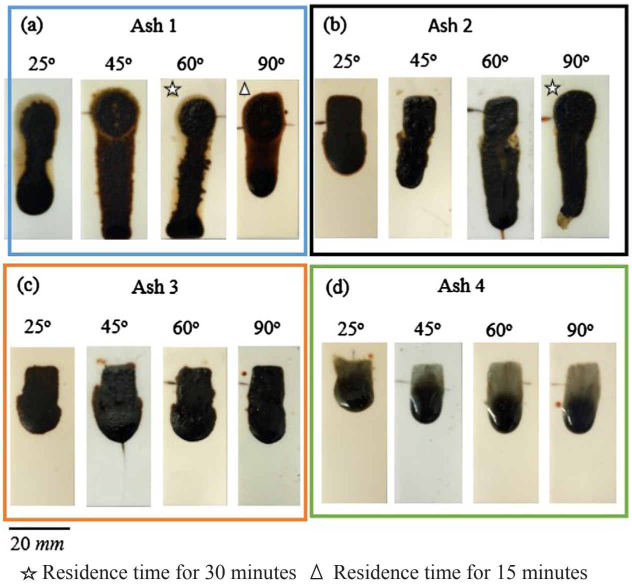


Fig. 9. Images for slag in different inclination angles. The temperature is 1400 °C and residence time is 40 min except label ones, (a) ash 1, (b) ash 2, (c) ash 3 and (d) ash 4.

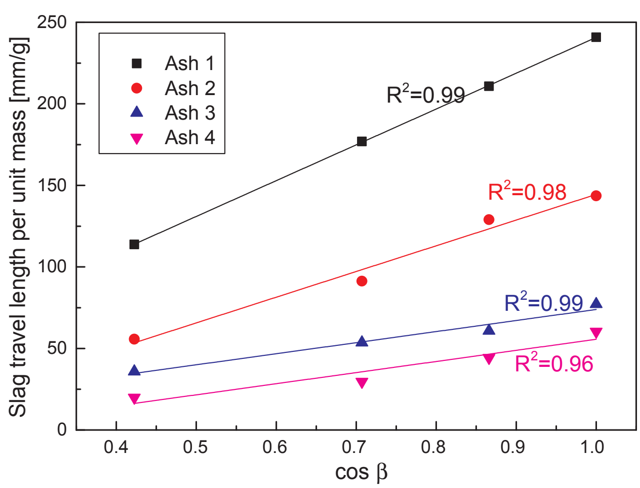


Fig. 10. The relationship between the inclined angle and slag travel length. The temperature is 1400 °C and residence time is 40 min.

$$ln\mu = Acos\beta + BlnL' + C \tag{16}$$

where $ln\mu$ can be deemed as the dependent variable, while $cos\beta$ and lnL' can be deemed as the independent variables. A, B and C are constants, which can be determined by the multiple regression fitting using all the experimental data achieved at 1400 °C in Fig. 12. The equation was obtained as the following Eq. (17).

$$ln\mu = 3.282281cos\beta - 1.882827lnL' + 7.397108 \tag{17}$$

Table 4 provides the regression statistics results for the multiple regression fitting, where the large correlation coefficient R^2 values indicate the reliability of the regression results. Additionally, it is essential to evaluate the residuals to determine whether they appear to fit the assumption of a normal distribution. A normal quantile plot of the standardised residuals, also known as Q-Q plot has been plotted and shown in Fig. 13. It is confirmed that all the datum points follow a linear trend, and the residuals do not seem to deviate from a random sample from a normal distribution in any systematic manner. This also validates the reliability of the regression results. Additionally, comparison was made between the experimentally measured trave length

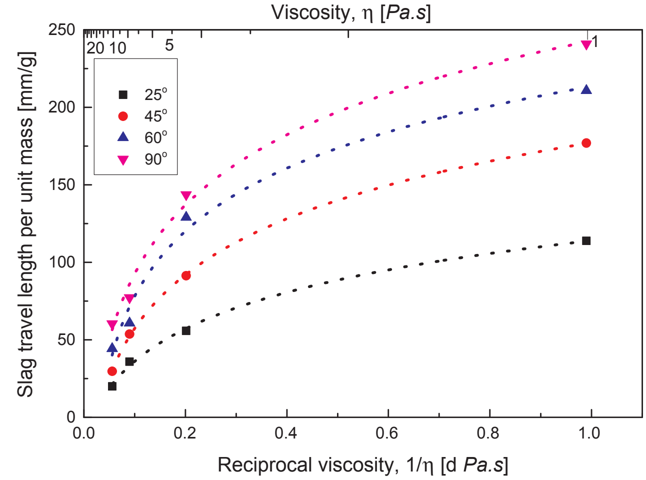


Fig. 11. The power law relationship between the viscosity and slag travel length, where the temperature is 1400 °C and exposure time is 40 min.

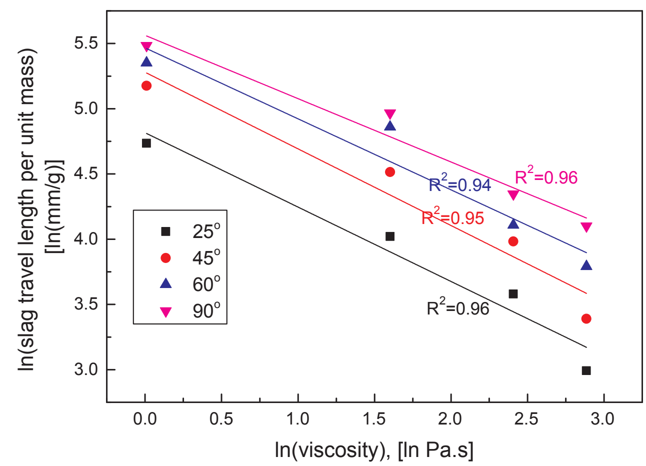


Fig. 12. The linear fitting for slag viscosity and travel length, where the temperature is 1400 °C and exposure time is 40 min.

and the respective value calculated based on Eq. (17). As evident in Fig. 14, almost all the predicted results are very close to the measured ones. Although slight deviations are observed for the highly viscous ash 3 and ash 4, the errors are still within an acceptable range.

3.4. Application and limitation

It is clear that the established M-IP methodology in this study is useful and accurate for the high-temperature slag viscosity measurement and prediction. Additionally, it is simple, cost-effective, fast and can be conducted in any ash research laboratory in the world, requiring maximum 0.2 g ash that is far less than a minimum mass of 50 g as per the existing commercial viscometers [3,5]. The presence of un-molten solid or solidification of molten slag is not influential either, since the whole moving slag will be measured for its travel length. In other

Table 4
Multiple linear regression statistics results for Eq. (17).

Regression Statistics			Coefficients	Standard Error
Multiple R	0.98987	Intercept (C)	3.92873	0.11120
R Square	0.97984	Variable 1 (A)	1.74328	0.13353
Adjusted R Square	0.97674	Variable 2 (B)	-0.53112	0.00449
Standard Error	0.11486			
Observations	16			

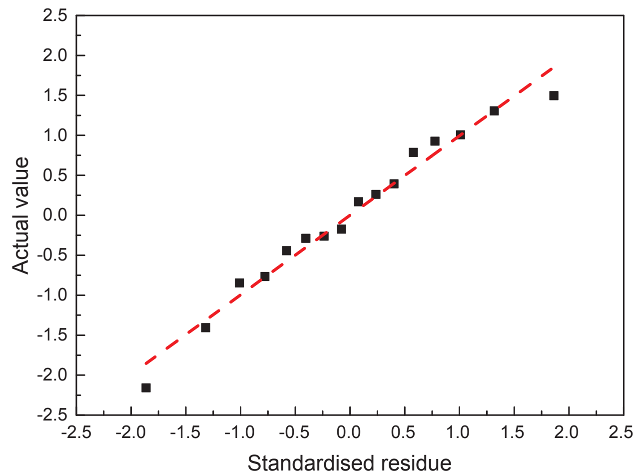


Fig. 13. Normal quantile plot for multiple linear regression.

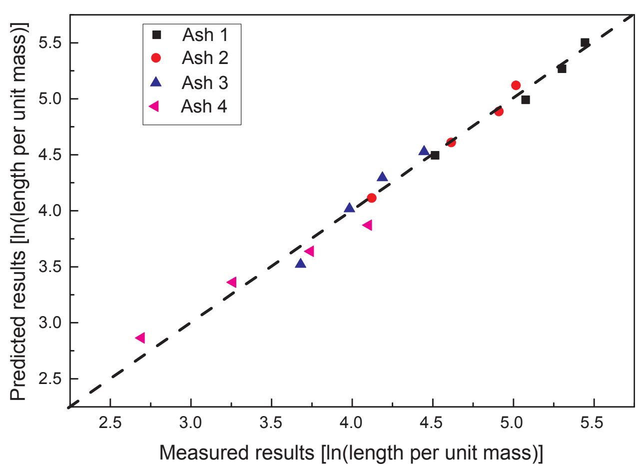


Fig. 14. The comparison of measurement and prediction results for standard ash samples 1–4.

words, the M-IP methodology developed here measure the 'apparent' viscosity of a slag, rather than only the viscosity of the molten liquid inside the slag. Moreover, by measuring only two temperature points, an Arrhenius equation can be established to predict the travel length at the intermediate temperatures. Subsequently, the resultant travel lengths can be plugged into Eq. (17) to predict the respective viscosities. The residence time for the measurement is recommended at 40 min for a considerably detectable travel length range for the viscosity calculation. Regarding the final empirical Eq. (17) established, it is clearly more objective than most of the existing equations that have to reply on the chemical composition of the slag. To iterate, most of the existing equation were developed based on the SiO_2-Al_2O_3 system [7–9], and hence, they are inapplicable to the highly alkaline ashes including the ash from brown coal, lignite and even those from the biomass. Such a problem should not exist for the methodology and its empirical equation developed here, since they are independent on the slag composition.

Due to the limitation of the horizontal furnace, the empirical Eq. (17) developed should only be employed up to 1400 °C. The prediction on any viscosity beyond 1400 °C using this equation is implausible, unless the higher temperatures were further tested using the same methodology. Additionally, the viscosity tested only covers a small range up to 17.9 *Pa·s* that is very close to the critical viscosity, 25 *Pa·s* for a continuous slag flow [30]. Therefore, any viscosity beyond this point may not be accurately measured by the inclined plane method, unless the higher temperatures (> 1500 °C) and different inclined plate are employed. Otherwise, it would rarely flow down even under an angle of 90°. Rather, the viscosity of the slag would only be qualitatively compared with the upper limit 17.9 *Pa·s* and the critical viscosity 25 *Pa·s*. However, from an engineering design perspective, this might be sufficient for a quick screening of the fuel to determine its suitability for entrained-flow gasifier or cyclone furnace. Finally, it is noteworthy that the interactions between corundum plate and slag could occur during the measurement, as that has been observed for the real ash samples in our previous works [18,19]. However, this may not affect the accuracy of the M-IP method developed here, because it was established based on the validation by prior testing synthetic standard ashes which have gone through the same/similar interactions as the real ashes would experience. In other words, for any new plates to be used, the standard synthetic ash samples should always be tested first. Future works will be conducted by us to further investigate these interactions in detail.

4. Conclusion

In this study, a simple and yet novel methodology based on the use of inclined plane, namely M-IP has been established and validated to determine the high-temperature slag viscosity. Seven synthetic standard ash samples have been tested on the inclined corundum plane, under the conditions of reducing environment of 1% CO in nitrogen, 1100 °C–1400 °C, 25°–90° and different duration time of 10–40 min. The slag mass was also varied from 100 *mg* to 400 *mg*. A multiple linear regression fitting was conducted to establish an empirical equation to predict slag viscosity (η) based on the slag travel length (L') and the inclined angle ($cos\beta$). The major conclusions can be drawn as follows:

1) The unit mass slag travel length is a function of slag mass, temperature and slagging exposure time. The slag travel length shows a linear relationship with mass within the mass range from 0.1 *g* to 0.3 *g* at 1400 °C. A linear relationship has been achieved between the logarithm of slag travel length and reciprocal of temperature, proving that the temperature dependence of the slag travel length (and viscosity) can be satisfactorily represented by the Arrhenius-type relationship.
2) Both the inclined angle and the reciprocal of slag viscosity affect the slag travel length in an intertwined manner, by following a linear relationship between the algorithm of slag travel length, cosβ and the logarithm of slag viscosity, as per the equation below:

$$ln\mu = 3.282281cos\beta - 1.882827lnL' + 7.397108$$

3) Both the large correlation coefficient value and the normal quantile plot validated the high accuracy of the above equation for the prediction of lag viscosity. The method is simple and only requires the tests on two different temperatures. However, due to the limitation of the slagging furnace, this empirical equation is limited to an upper temperature of 1400 °C and a maximum viscosity of 17.9 *Pa·s*. Any larger viscosity would only be measured qualitatively, although it is sufficient for a comparison with the critical viscosity 25 *Pa·s* from the perspective of a quick screening of solid fuel for entrained-flow gasifier or cyclone furnace.

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